

# Customer Churn Intelligence & Retention Platform

## Final Product Requirements Document

**Version:** 2.0   **Status:** MVP / Development Ready

**Product Type:** Machine Learning + Analytics + Explainable AI + Decision Support

**Core Product Promise:** Predict → Explain → Prioritize → Act

## 1. Executive Summary

The **Customer Churn Intelligence & Retention Platform** is a machine-learning-powered decision-support product that helps businesses identify customers who are at risk of leaving, understand the reasons behind that risk, prioritize intervention, and recommend appropriate retention actions.

The product transforms:

Customer Data → Churn Prediction → Churn Probability → Risk Classification → Explanation → Recommended Action

The supplied project dataset contains **64,374 customer records and 12 columns**, with **Churn** as the target variable. CustomerID must be excluded from model training.

## 2. Problem Statement

Businesses frequently discover churn after the customer has already decided to leave. Traditional reporting can answer how many customers churned, but a useful retention system must answer: **Which customers are likely to churn next, how likely are they to churn, why are they at risk, and what should we do about them?**

- Retention teams cannot efficiently prioritize customers.
- High-risk customers may be identified too late.
- Support problems may remain unresolved.
- Payment difficulties may be overlooked.
- Low-engagement customers may silently disappear.
- Retention campaigns may be applied without meaningful risk signals.

## 3. Product Vision

**Build an explainable customer-retention intelligence platform that identifies churn risk early, explains the major drivers behind that risk, prioritizes customers by business importance, and recommends targeted actions before valuable customers leave.**

The product should not simply predict churn. It should support a complete business decision: **Who → Why → How Urgent → What Next.**

## 4. Product Goals

**G1 — Predict Churn** — Predict whether a customer is likely to churn.

**G2 — Quantify Risk** — Return a probability rather than only a Yes/No prediction.

**G3 — Prioritize Customers** — Classify customers into Low, Medium, and High Risk.

**G4 — Explain Predictions** — Identify the major factors contributing to the customer's predicted risk.

**G5 — Recommend Retention Actions** — Translate risk factors into practical retention recommendations.

**G6 — Provide Business Intelligence** — Help managers understand overall churn, churn drivers, customer segments, and retention opportunities.

**G7 — Enable Scalable Scoring** — Support both individual customer prediction and batch customer-risk scoring.

## 5. Product Scope

### MVP

- Dataset upload
- Data validation
- Data preprocessing
- Exploratory analytics
- Four ML models
- Model comparison
- Model selection
- Hyperparameter tuning
- Churn prediction
- Churn probability
- Risk classification
- Explainability
- Retention recommendations
- Analytics dashboard
- Customer risk profile

### Post-MVP

- Batch scoring
- Customer risk queue
- Model monitoring
- Automated alerts
- CRM integration
- Retention campaign tracking
- Revenue-at-risk estimation
- Continuous retraining

## 6. Target Users

**Retention Manager:** Who is likely to churn? Which customers should be contacted first? Why are they at risk? What action should be taken?

**Customer Support Manager:** High-risk customers, repeated support-call customers, and customers requiring escalation.

**Business Manager:** Overall churn rate, risk distribution, major churn drivers, retention opportunities, and business impact.

**Data Scientist / ML Engineer:** Dataset analysis, model comparison, model performance, feature importance, hyperparameter tuning, and model versioning.

## 7. Dataset Requirements

| Feature           | Type        | Purpose                        |
|-------------------|-------------|--------------------------------|
| Age               | Numerical   | Demographic context            |
| Gender            | Categorical | Demographic analysis           |
| Tenure            | Numerical   | Customer relationship duration |
| Usage Frequency   | Numerical   | Engagement                     |
| Support Calls     | Numerical   | Service interaction            |
| Payment Delay     | Numerical   | Payment behavior               |
| Subscription Type | Categorical | Product plan                   |
| Contract Length   | Categorical | Contract commitment            |
| Total Spend       | Numerical   | Customer value                 |
| Last Interaction  | Numerical   | Interaction behavior           |

**Identifier:** CustomerID — must be excluded from predictive features.

**Target:** Churn — 0 = customer stayed; 1 = customer churned.

## 8. Data Quality Requirements

- Required columns
- Row and column counts
- Data types
- Missing values
- Duplicate records
- Invalid numerical values
- Invalid categorical values
- Feature ranges
- Outliers
- Target distribution

The system must validate rather than silently assume that future datasets have the same quality as the current dataset.

## 9. Data Preprocessing

- Remove CustomerID.
- Separate Churn from input features.
- Identify numerical and categorical features.
- Handle missing values when present.
- Validate categorical values.
- Encode categorical variables.
- Scale numerical features where required.
- Preserve the exact preprocessing pipeline for prediction.

## 10. Exploratory Data Analysis

- Age vs Churn

- Tenure vs Churn
- Usage Frequency vs Churn
- Support Calls vs Churn
- Payment Delay vs Churn
- Total Spend vs Churn
- Last Interaction vs Churn
- Gender vs Churn
- Subscription Type vs Churn
- Contract Length vs Churn
- Correlation matrix
- Feature distributions and comparisons

## 11. Dataset-Derived Product Intelligence

The uploaded analysis identifies several useful patterns in this particular dataset. These observations are **dataset-specific** and must not be treated as universal churn rules.

- **Payment Delay:** reported as the strongest numerical relationship with churn, approximately +0.557 correlation.
- **Support Calls:** positive relationship with churn, approximately +0.305 correlation.
- **Usage Frequency:** negative relationship with churn, approximately -0.115 correlation.
- **Tenure:** positive relationship with churn in this dataset, approximately +0.195 correlation.
- **Contract Length:** monthly customers reportedly show higher churn than quarterly and annual customers.
- **Subscription Type:** differences are reported as smaller than those associated with payment delay and support calls.
- **Last Interaction:** reported as having approximately zero correlation with churn.

## 12. Correlation and Feature Analysis

- Generate a correlation matrix.
- Identify numerical features associated with churn.
- Identify relationships between input variables.
- Detect potentially redundant features.
- Compare correlation findings against EDA.
- Compare correlation findings against model-based feature importance.

Correlation must not be treated as proof of causation.

## 13. Train/Test Strategy

**80% training / 20% testing** with **random\_state = 42** and stratification to preserve churn-class distribution. The test dataset must remain unseen during model fitting.

## 14. Machine Learning Models

**Logistic Regression** — Strong baseline and interpretable classification.

**Random Forest** — Non-linear classification and feature-importance analysis.

**Support Vector Machine** — Alternative classification approach using scaled features.

**K-Nearest Neighbors** — Distance-based classification comparison.

## 15. Model Evaluation

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Optional: ROC-AUC

**Primary business consideration:** recall for the churn class is important because a false negative represents a customer who actually churns but was not identified. However, recall should not be maximized blindly; excessive false positives can create unnecessary retention costs.

## 16. Model Selection Strategy

- Churn-class recall
- F1-score
- Precision
- Accuracy
- ROC-AUC where appropriate
- Cross-validation stability
- Probability calibration
- Interpretability
- Business usefulness

The final model must not be predetermined. Selection should balance predictive performance with the business cost of errors.

## 17. Confusion Matrix

| Actual / Predicted | Stay           | Churn          |
|--------------------|----------------|----------------|
| Stay               | True Negative  | False Positive |
| Churn              | False Negative | True Positive  |

The system shall explain which errors are most costly for the business.

## 18. Hyperparameter Optimization

- Use GridSearchCV or equivalent cross-validation.
- Record best parameters.
- Record best cross-validation score.
- Compare baseline and tuned test scores.
- Record improvement.
- Evaluate the tuned model on unseen test data.

## 19. Probability Calibration

Because churn probability is a first-class product output, the final model should be checked for probability calibration where appropriate. An output such as **Churn Probability: 82%** should represent a meaningful estimate of risk rather than

merely a model confidence score.

## 20. Churn Prediction Engine

For a new customer, accept: Age, Gender, Tenure, Usage Frequency, Support Calls, Payment Delay, Subscription Type, Contract Length, Total Spend, and Last Interaction.

Return: **Churn Prediction, Churn Probability, Risk Level, Risk Drivers, and Recommended Action.**

## 21. Risk Engine

The product shall classify customers into Low, Medium, and High Risk.

Risk thresholds must be **configurable**. Initial thresholds may be used for MVP demonstration, but final thresholds should eventually be determined using model calibration, historical outcomes, retention cost, customer value, and the cost of false positives versus false negatives.

## 22. Explainability

The platform must answer: **Why does this customer have a high churn probability?**

### Global Explainability

Shows what generally influences churn predictions, such as Payment Delay, Support Calls, Tenure, Usage Frequency, Contract Length, and Total Spend.

### Individual Explainability

Shows why a particular customer is at risk, for example: high payment delay, frequent support calls, low usage frequency, or a short contract.

**Global model importance  $\neq$  individual customer explanation.**

## 23. Retention Recommendation Engine

**High Payment Delay** — Investigate payment difficulties, resolve payment issues, and provide appropriate payment support.

**Frequent Support Calls** — Review support history, escalate unresolved issues, and assign proactive support.

**Low Usage** — Use product education, personalized engagement, or usage campaigns.

**Monthly Contract** — Evaluate whether a longer-term plan is appropriate and present suitable offers where commercially justified.

**High Overall Risk** — Prioritize the customer for direct retention intervention.

Recommendations are decision-support suggestions, not automatic actions.

## 24. Customer Risk Profile

- Customer information
- Behavioral information
- Churn prediction
- Churn probability
- Risk level
- Top risk drivers
- Recommended intervention
- Priority

## 25. Executive Dashboard

- Total Customers
- Churned Customers
- Retained Customers
- Churn Rate
- High-Risk Customers
- Medium-Risk Customers
- Low-Risk Customers
- Top churn drivers
- Churn by contract
- Churn by subscription
- Churn by usage
- Churn by payment behavior
- Churn by support behavior

## 26. Customer Risk Queue

The retention manager should see a prioritized customer list containing Customer, Churn Probability, Risk, Primary Driver, and Recommended Action. Actual values must be generated dynamically.

## 27. Batch Prediction

The platform should support CSV batch scoring. Input is a customer dataset without the Churn target. Output should contain CustomerID, Churn Prediction, Churn Probability, Risk Level, Major Risk Drivers, and Recommended Action.

## 28. Analytics Dashboard

- Churned vs Retained
- Payment delay vs churn
- Support calls vs churn
- Usage frequency vs churn
- Tenure vs churn
- Monthly vs Quarterly vs Annual
- Basic vs Standard vs Premium
- Feature importance
- Model comparison
- Confusion matrix

## 29. Functional Requirements

| ID    | Requirement        | Description                                                                      |
|-------|--------------------|----------------------------------------------------------------------------------|
| FR-01 | Dataset Upload     | Accept the supported customer dataset format.                                    |
| FR-02 | Data Validation    | Validate required fields, types, missing values, duplicates, and invalid values. |
| FR-03 | Data Preprocessing | Clean, encode, and scale data as required.                                       |

| ID    | Requirement               | Description                                                   |
|-------|---------------------------|---------------------------------------------------------------|
| FR-04 | Feature/Target Separation | Churn shall never be included in model inputs.                |
| FR-05 | Model Training            | Train the four required baseline models.                      |
| FR-06 | Model Evaluation          | Calculate required classification metrics.                    |
| FR-07 | Model Selection           | Identify the best model using predefined evaluation criteria. |
| FR-08 | Hyperparameter Tuning     | Optimize the selected model.                                  |
| FR-09 | Individual Prediction     | Predict churn for a new customer.                             |
| FR-10 | Probability               | Return churn probability.                                     |
| FR-11 | Risk Classification       | Assign configurable risk level.                               |
| FR-12 | Explainability            | Provide customer-level risk drivers.                          |
| FR-13 | Recommendations           | Generate retention recommendations.                           |
| FR-14 | Batch Prediction          | Support bulk customer scoring.                                |
| FR-15 | Analytics                 | Provide churn and customer-behavior analytics.                |
| FR-16 | Model Monitoring          | Future versions should monitor performance and data drift.    |

## 30. Non-Functional Requirements

- **Performance:** individual prediction should return within approximately 2 seconds after model load.
- **Reliability:** training and prediction must use identical preprocessing logic.
- **Security:** customer information must be accessible only to authorized users.
- **Explainability:** predictions must be understandable to non-technical users.
- **Scalability:** architecture should support thousands to millions of customer records as the product evolves.
- **Maintainability:** ML pipeline should be modular.

## 31. Recommended Technology Stack

- **Machine Learning:** Python, Pandas, NumPy, Scikit-learn
- **Visualization:** Plotly, Matplotlib, Seaborn
- **Backend:** FastAPI
- **Frontend:** React for production; Streamlit for a rapid academic MVP
- **Storage:** CSV/SQLite for MVP; PostgreSQL for production
- **Deployment:** Docker and cloud deployment

## 32. Recommended Application Structure

1. Executive Dashboard
2. Customer Risk Queue
3. Customer Profile
4. Predict Customer
5. Churn Analytics
6. Model Performance
7. Explainability
8. Batch Scoring
9. Model Management

## 33. End-to-End User Journey



LOGIN  
↓  
EXECUTIVE DASHBOARD  
↓  
VIEW CHURN / RISK OVERVIEW  
↓  
OPEN CUSTOMER RISK QUEUE  
↓  
SELECT CUSTOMER  
↓  
VIEW CHURN PROBABILITY  
↓  
VIEW RISK LEVEL  
↓  
VIEW RISK DRIVERS  
↓  
VIEW RECOMMENDED ACTION  
↓  
TAKE RETENTION ACTION  
↓  
TRACK OUTCOME

## 34. Business Success Metrics

### ML Metrics

- Recall
- Precision
- F1-score
- Accuracy
- ROC-AUC
- Probability calibration

### Business Metrics

- High-risk customers identified
- Retention intervention rate
- Intervention success rate
- Customers retained after intervention
- Churn reduction
- Revenue retained
- Cost per successful intervention
- False-positive intervention cost
- False-negative opportunity cost

## 35. Product Safety and Limitations

The platform must clearly communicate that a churn prediction is an **estimate, not a guarantee**. A probability means that the customer's historical characteristics resemble patterns associated with churn; it does not mean the customer will definitely leave.

The system is therefore a **decision-support tool**. Retention recommendations must remain subject to human review.

## 36. Future Roadmap

## Phase 1 — MVP

- Dataset ingestion
- Data validation
- EDA
- Four ML models
- Evaluation
- Model selection
- Hyperparameter tuning
- Individual prediction
- Risk classification
- Explainability
- Recommendations

## Phase 2 — Operational Retention

- Batch scoring
- Customer risk queue
- Automated alerts
- Customer segmentation
- Model monitoring
- Prediction history

## Phase 3 — Business Integration

- CRM integration
- Customer-success workflows
- Retention campaign tracking
- Revenue-at-risk
- Customer lifetime value

## Phase 4 — Intelligent Retention

- Continuous retraining
- Drift detection
- Real-time scoring
- Personalized interventions
- A/B testing of retention strategies

## 37. MVP Acceptance Criteria

### Dataset

- ✓ Dataset loads successfully.
- ✓ 64,374 customer records can be processed.
- ✓ 12 columns are correctly recognized.
- ✓ CustomerID is excluded from model features.

- ✓ Churn is separated as the target.
- ✓ Missing values and duplicates are checked.
- ✓ Data types and categorical values are validated.

## Preprocessing

- ✓ Numerical features are processed correctly.
- ✓ Categorical features are encoded.
- ✓ Scaling is applied where required.
- ✓ Training and prediction use the same preprocessing pipeline.

## Machine Learning

- ✓ All four required models are trained.
- ✓ 80/20 split is implemented.
- ✓ random\_state=42 is used.
- ✓ Stratification is implemented.
- ✓ Accuracy, precision, recall and F1-score are calculated.
- ✓ Confusion matrix is generated.
- ✓ Feature importance is analyzed.
- ✓ Best model is selected.
- ✓ Hyperparameter tuning is completed.

## Prediction

- ✓ New customer information can be entered.
- ✓ Churn prediction is returned.
- ✓ Churn probability is returned.
- ✓ Risk level is returned.
- ✓ Risk drivers are displayed.
- ✓ Recommended action is displayed.

## Product

- ✓ Executive dashboard works.
- ✓ Customer risk profile works.
- ✓ Analytics dashboard works.
- ✓ Model-performance dashboard works.
- ✓ High-risk customers can be identified.
- ✓ Batch prediction is supported or clearly scoped for the next release.

## 38. Key Product Differentiator

The system should not be presented as **“a machine learning model that predicts customer churn.”**

It should be presented as an **“explainable customer churn intelligence and retention platform.”**

A prediction model answers: **Will they churn?**

This product answers: **Will they churn? How likely? Why? How urgent? What should we do?**

## 39. Final Product Definition

### Customer Churn Intelligence & Retention Platform

A machine-learning-powered decision-support platform that analyzes customer behavior, predicts churn probability, classifies customer risk, explains the key drivers behind that risk, prioritizes customers for intervention, and recommends targeted retention actions.

#### Product Equation

CUSTOMER DATA  
+  
MACHINE LEARNING  
+  
RISK ANALYSIS  
+  
EXPLAINABILITY  
+  
RETENTION STRATEGY  
=  
CUSTOMER RETENTION INTELLIGENCE

#### Final Product Flow

Customer Data  
↓  
Data Validation  
↓  
Preprocessing  
↓  
ML Prediction  
↓  
Churn Probability  
↓  
Risk Classifier  
↓  
Explanation  
↓  
Customer Prioritization  
↓  
Retention Recommendation  
↓  
Human Action  
↓  
Retention Outcome  
↓  
Model / Business Feedback

**Final positioning:** Predict → Explain → Prioritize → Act → Learn